

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/355872171>

Crime Prediction and Forecasting using Machine Learning Algorithms

Article in *International Journal of Computer Science and Information Technology* · March 2021

CITATIONS

4

READS

2,415

5 authors, including:



[Azwad Tamir](#)

University of Central Florida

5 PUBLICATIONS 65 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



EEE Department Development Project [View project](#)



3D IC Design [View project](#)

Crime Prediction and Forecasting using Machine Learning Algorithms

Azwad Tamir^{#1}, Eric Watson[#], Brandon Willett[#], Qutaiba Hasan[#], Jiann-Shiun Yuan^{#2}

[#]Department of ECE, University of Central Florida
4000 Central Florida Blvd, Orlando, FL 32816, USA

¹a.tamir@Knights.ucf.edu

²Jiann-Shiun.Yuan@ucf.edu

Abstract— This research will focus on machine learning algorithms for crime forecasting. In the modern world, crime is becoming a major and complex problem. In this research, we discover the best course of action for teaching a model to forecast crime in major metropolitan cities. The purpose of this study is to provide the Police Department with proper crime forecasting so they can better delegate their resources in response to future crime hotspots. We applied several machine learning models to predict the severity of a reported crime based on whether the crime would lead to an arrest or not. We also did a deep dive into the city districts and studied the crime trends by year. We used Folium to do data visualization for the study of these trends. We discovered trends in the number of crimes and the arrest rate from year to year. The different machine learning models that we developed are the Random Forest, K-Nearest-Neighbours, AdaBoost, and Neural Network. We tested our models on the Chicago Police Department's CLEAR (Citizen Law Enforcement Analysis and Reporting) system, which has more than 6,000,000 records. Among all four models, the Neural Network has the best outcome with an accuracy of 90.77%. This study also provides an insight into the applicability of different machine learning models in analyzing crime report datasets from large metropolitan cities.

Keywords— AdaBoost, crime forecasting, deep neural network, folium, future crime, KNN, machine learning, prediction, random forest.

I. INTRODUCTION

Aggressive urbanization is causing a rapid increase in the size and population of large cities across the world. This massive population growth in cities is causing total crimes to rise sharply, making it difficult for the police department to keep up with it. It is not possible for the police department to station police officers in every street corner, so they need some intelligent system that can predict the likelihood of crimes occurring in different parts of the city at different times of the day. Also, there is a large volume of emergency 911 calls and crime reports coming at every moment of the day. These reports need to be sorted to identify the more imminent threats, so the police department could allocate their resources accordingly and respond to more alarming situations before addressing the rest.

These difficulties stimulated a lot of research work in recent times related to predicting future crimes to help the police department allocate their resources. S. Kim et al. proposed different machine learning methods to predict

crimes in Vancouver from crime data in the last 15 years and obtained an accuracy of 39% and 44% for the K-nearest-neighbour and boosted decision tree algorithms, respectively [1]. Y. Lin et al. investigated a deep machine learning model based on the broken windows and spatial analysis to predict future crime hotspot locations in Taiwan [2]. P. Kumari et al. applied Extra Tree Classifier, K-Neighbours, Support Vector Machines, Decision Tree Classifier, and Neural Network algorithms to predict the probability of different types of crimes in different locations and time around the city [3]. N. Hooda et al. used data obtained from different government websites to predict future crime rates at different city areas using the Folium API [4]. W. Gorr et al. analysed crime data to influence and adjust policies and rules to better adapt for the future [5]. E. Ahishakiye et al. came up with a machine learning model using decision trees to categorize crimes based on the crimes' text description [7]. H. Nguyen et al. build an automatic machine learning based classifier that predicts the severity of a traffic accident based on the incident records [8]. J. Cohen et al. proposed a model for crime forecasting to help with the allocation of police resources [9]. Lastly, A. Wheeler et al. used Random Forest to build a long-term crime predictor of different locations in the city of Dallas.

The objective of our work is to predict the severity of the crime based on the incident reports. We choose whether an arrest was made as the indicator representing the crime severity. We used a dataset of crime reports for the city of Chicago from the year 2001 to 2018, which was made available through the Kaggle website [13]. The data came from the Chicago Police Department's CLEAR (Citizen Law Enforcement Analysis and Reporting) system and consisted of more than 6 million records or data points. Each of the data point has 23 features that represented various information about the nature, location, time, description, and severity of the reported crime. We generated four different machine learning models, namely Random forest, K-nearest-neighbours (KNN), Adaptive Boost (AdaBoost), and artificial neural network (NN) to predict whether an arrest was made based on the information available in the other features of the dataset. Moreover, we generated detailed analyses relating to the crimes' location and trends and visualized them in interactive maps using the Folium API.

The paper is organized into four subsequent sections. Section II contains all the data pre-processing and feature engineering steps taken to clean up the data and prepare it for

implementation. Section III consists of the details of the model architecture and properties and parameters used to construct the classifiers. Section IV incorporates the details of the analysis done on the data and the techniques used to visualize the trends and crime patterns. Section V reports the results of the various models and demonstrates the visualizations. Finally, Section VI concludes the work and points out potential future research directions.

II. DATA PRE-PROCESSING & FEATURE SELECTION

A series of pre-processing steps were implemented on the dataset to clean it up and make it ready before inserting them into the machine learning classifiers. The detailed analysis of the preprocessing steps taken for each of the features is provided below:

ID/Case Number: Both the ID and Case Number features are unique identifiers used to tag each incident or datapoint. It had unique values for each of the data points and hence did not contain any information. These features, however, were used to remove duplicate data from the dataset. After this, they were not considered and entirely dropped from the dataset.

Block: This represents the block address of the reported crimes. There were two parts to this feature. The first was a set of numbers that denotes the house number of the report location. The second part consisted of the name of the street of the reported location. There were several types of inconsistencies in this feature that were first removed. Some of these inconsistencies include typos, different terminologies used for missing datasets, and formatting errors. There were a few missing values for which the entire row was discarded. There were too few of them to cause any significant decrease in the number of datapoints. The next modification was to separate the number and street name into two different features. This was done in order to make it easier for the training models to learn the dataset. Finally, the street names were encoded into integers for the convenience of storing the values and loading them into the machine learning models.

Description: This contains the description of a crime in a concise manner. There were 353 unique description values in the dataset. As a result, it consists of categorical information about the nature of the reported crime. The description texts were encoded into integers to help with the storage and loading process. The feature data were also checked for missing values, typos, inconsistencies, and outliers, but none were found.

IUCR and FBI code: IUCR stands for Illinois Uniform Crime Reporting, and this was used as a code denoting the type of crimes reported. The FBI code also contains similar information about the nature of the crime. These two features contain information that is very similar to the description feature, and there also exists a very high correlation among them. Hence, these features were dropped as this information is already being feed to the machine learning models through the description column.

Arrest: This is a binary categorical data that has either true or false values. True means that an arrest was made for the

reported crime while false refers that no one was arrested for that crime. This represents the severity of the crime and could be used to deduce whether imminent police involvement is required or not. The police department could use this information to control the allocation of resources and assign police officers accordingly. So, this was used as the prediction labels of the machine learning models. The data was also encoded into binary values for ease of handling.

Domestic: This also consists of binary values and denotes whether the crime occurred in a domestic or public environment. Domestic crimes are usually more non-violent and include domestic disturbance and other related issues. The data was first encoded into integers and checked for inconsistencies and missing values. Fortunately, there was no problem with the data and required no further processing.

District, Ward, and Community Area: These are three different measures used to divide up the total area under a city. They contained the location information of the crime report. However, all of them had a large amount of missing data points and there exists other location features like the coordinates and Block which contained very little missing values and the same information. Hence, all these features were excluded from the training process.

Beat: This represents the smallest police geographical segmentation area, which they use to allocate resources. This feature was already in the form of integers and contained no missing values, inconsistencies, and outliers. As a result, no further processing was required.

Latitude and Longitude: These contained the exact location of the crime's reporting location in the form of continuous variables. About 1 percent of the data was missing. The entire row containing the missing values were excluded from the dataset. The data also contained some inconsistencies and outliers. These were dropped as well.

X and Y coordinate: Contains the same information as the latitude and longitude values in a different format. Hence, they contained no additional information and were therefore dropped.

Updated on: This represents when the report was included in the police server. They show a high correlation to the data and time features of the dataset and hence were dropped.

Location: Composite of the latitude and longitude columns, they contained the same information as the latitude and longitude features and were therefore not included.

Location Description: This is a categorical data that describes the location of where the crime was committed. There are over 170 unique values, which were encoded into integers to make storing the values easier and loading them into the machine learning model. Over 1,968 null values were found for this feature, which were dropped from the dataset.

Primary Type: This is a categorical data that describes the type of crime that was committed. There were over 35 unique values of this data, which were encoded into integers for making it more convenient to store values and load them into the machine learning model. Some unique values also had lower value counts than others, so values with a count less

than 1,000 were removed from the dataset, leaving only 25 unique values for this data.

Date: This data contains the time at which the crime incident was reported. This feature was used as an index when pre-processing the dataset and was dropped after it was not needed.

Year: This feature contains the year at which the crime was reported. The unique values are the years of 2001 to 2017. When analyzing the data, shown in the heatmap in Fig. 1, the year 2017 only contained crimes committed in January. Data containing the year of 2017 was dropped entirely from the dataset. This value was also encoded for the dataset, such as the value of 0 representing 2001.

Month: This feature contains the month at which the crime was reported. The date feature of the crime reported was used to generate it. This feature was also encoded after it was created to make it easier to store and load them into the machine learning model.

samples to split an internal node was set to 2; minimum samples per leaf were set to 1; and bootstrap sampling was set to true. Five-fold cross-validation was used to train and evaluate the random forest model, which gave the average output scores shown in Table 5.

In order to improve the overall scores of the random forest classifier, the random search and grid search functions from the Scikit-learn library were used to improve the model for arrest prediction. Before using the random search function, 10% of the Chicago Police dataset was uniformly sampled for training and evaluation, which was used to help reduce training time. The random search function utilized five-fold cross-validation for evaluating the model, which also took a set of parameters chosen for the random forest model, shown in Table 1. The random search was set with a max of 100 iterations, where each iteration took random parameters to train and evaluate from the set. After the random search was performed, the parameters from the best model were chosen,

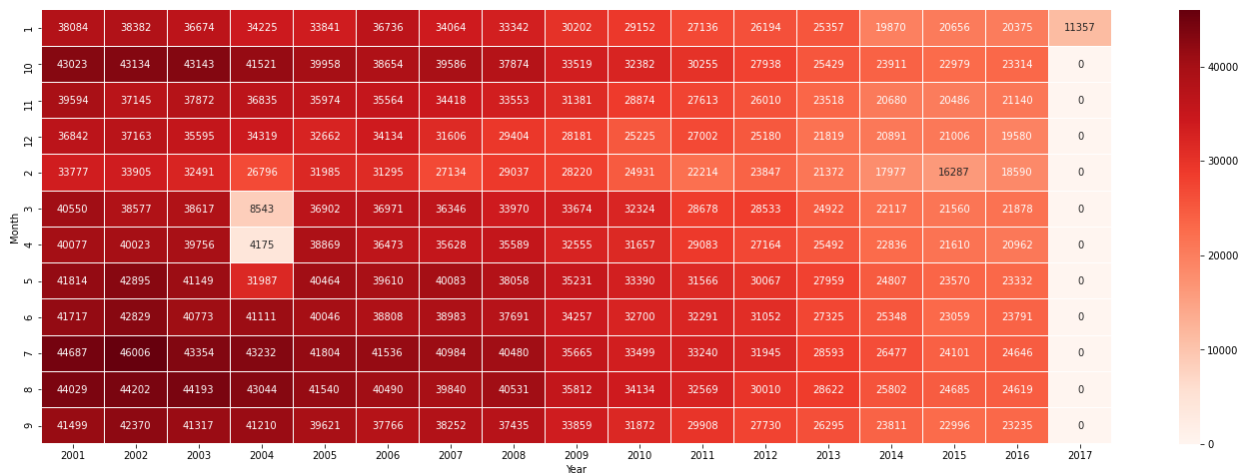


Fig. 1 Heatmap showing the number of crimes committed per month for all years

Weekday: This feature contains the weekday on which the crime was reported. Like the month feature, the date feature of the crime report was used to generate this feature.

After adjusting all the above individual feature pre-processing, there were in total 14 features chosen for training the classification machine learning models to predict the arrest status.

III. MODELS

Several different machine learning models were applied to the dataset. The detailed architecture of the models and their parameters are discussed in this section.

A. Random Forest

The random forest classifier was the first machine learning model applied on the Chicago Police dataset to predict arrests. The Scikit-learn machine learning library [11] was used to implement the random forest classifier on the pre-processed dataset. For initial training and evaluation, the random forest's default parameters were used, which included 100 decision trees. Gini was used as the impurity measurement; minimum

which was used to perform a grid search using the following parameters shown in Table 2. Compared to the random search, the grid search accounts for all parameter combinations. After performing a grid search, the best number of estimators was 550, and the best max depth of each tree was 20. Next, 75% of the Chicago Police dataset was uniformly sampled for training,

TABLE I
RANDOM SEARCH PARAMETERS FOR RANDOM FOREST MODEL

Parameter	Parameter Values
Number of Estimators	[100, 200, 300, 400, 500, 600, 700, 800, 900, 1000]
Max Depth of the Tree	[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
Minimum Samples to Split an Internal Node	[2, 4, 8]
Minimum Samples Required at a Leaf Node	[1, 2, 4]
Max Features when Considering a Best Split	['sqrt', 'log2']
Bootstrap Samples when Building a Tree	[True, False]

while the remaining 25% were used for the test set. Compared to the scores using the default parameters, the accuracy increased around 3.98%, precision improved around 11.88%, recall increased around 2.46%, and the f1 score increased around 6.66%. The performance results of the tuned random forest model are shown in Table 5.

TABLE III
GRID SEARCH PARAMETERS FOR RANDOM FOREST MODELS

Parameter	Parameter Values
Number of Estimators	[450, 500, 550]
Max Depth of the Tree	[15, 20, 25]
Minimum Samples to Split an Internal Node	[2]
Minimum Samples Required at a Leaf Node	[2]
Max Features when Considering a Best Split	['log2']
Bootstrap Samples when Building a Tree	[False]

B. K-Nearest Neighbours

The k-nearest neighbors classifier was the second machine learning model used on the Chicago Police dataset to predict arrests. Like the random forest classifier, the Scikit-learn library was used to implement the k-NN classifier on the dataset. For initial training and evaluation, the default parameters of the k-NN classifier were used, with a K value equal to 5 and the distance measurement being set as Euclidean. The Five-fold cross-validation was used to train and evaluate the k-NN model, which gave the average output scores shown in Table 5.

In order to improve the k-NN model, the random search function from Scikit-learn was used to find the best parameters for the model, where the parameters chosen are shown in Table 3. Like the random forest model, 10% of the dataset was uniformly sampled for training/evaluating the model, where 100 iterations were used for the random search along with the five-fold cross-validation. The best model gave the following parameters: k = 11, weights = 'distance', algorithm = 'ball_tree', leaf_size = 20, and p = 1. A grid search was not further needed for this model, so the parameters from the best model in the random search were

TABLE III
RANDOM SEARCH PARAMETERS FOR KNN MODEL

Parameter	Parameter Values
Number of Neighbors	[3, 5, 7, 9, 11, 13]
Weights (Function used in Prediction)	['uniform', 'distance']
Algorithm (Used for nearest neighbors computation)	['ball_tree', 'kd_tree']
Leaf Size (Used for BallTree or KDTree)	[20, 30, 40]
Power Parameter (Used for Minkowski Metric)	[1, 2]

used for training and evaluating the model. Like the random forest model, 75% of the dataset was uniformly sampled, and the remaining was used for the test set. The performance of the tuned k-NN model is shown in Table 5. The k-NN accuracy increased around 0.94%, precision increased around 4.07%, recall decreased around 0.49%, and f1 score increased around 0.41%. Overall, compared to the random forest model, the k-NN model did not show significant improvement apart from a slight increase in precision.

TABLE IV
MODEL PROPERTIES AND HYPERPARAMETERS OF NN MODEL

Parameter Description	Value
Loss Function	Cross Entropy Loss
Optimizer	Adam Optimizer
Test Train Split	Fivefold cross validation
Output Class	2
Batch Size	1000
Learning Rate	0.01
Number of Training Epochs	500

C. AdaBoost

The adaptive boost classifier was the third machine learning model used on the Chicago Police dataset to predict arrests. Like the previous two models, the Scikit-learn library was used to implement the adaptive boost classifier on the dataset. For training and evaluating the classifier, five-fold cross-validation was used and the default parameters of the AdaBoost model were used. The default parameters include a learning rate of 1; 200 weak classifiers; and a max depth of 1 for each decision tree classifier. The model's performance with the default parameters is shown in Table 5, which was shown to be lower in performance than the random forest model, which is more likely due to it being more susceptible to noise in the data.

Like the previous two classifiers, the Scikit-learn functions were used to help tune the model. Given that the only important parameters to tune were the number of estimators and the learning rate, a grid search technique was used as there were fewer parameter combinations to work with compared to the other models. The parameters chosen for the estimators were: 400, 500, and 600. Also, learning rates of 1, 1.2, 1.4, 1.6, 1.8, and 2 were considered. Like the previous models, 10% of the dataset was uniformly sampled for training and evaluating the model, where five-fold cross-validation was used for the grid search. The parameters from the best model were a number of estimators equal to 600, and a learning rate equal to 1.8. After the best parameters were found, 75% of the dataset was uniformly sampled for training and the remaining 25% was used for the test set to evaluate the model. The performance of the tuned AdaBoost model is shown in Table 5. The accuracy of the model increased around 2.05%, precision decreased around 0.82%, recall increased around 8.55%, and the f1 score increased around

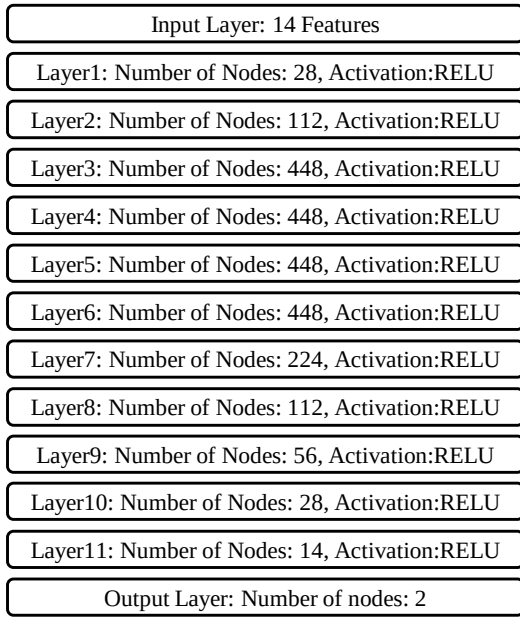


Fig. 2 The detailed architecture of the best performing neural network model

6.31%. Overall, this led to the AdaBoost classifier performing best after the random forest classifier, which has the k-NN classifier perform below the AdaBoost classifier.

D. Neural Network

The last classification machine learning model that we trained on the Chicago Police dataset to predict arrest was a neural network based classifier. Four different models that varied in architecture and number of layers were trained on the dataset, and their test accuracies were compared. The first model consisted of 4 hidden layers, the second model had 6 hidden layers, whereas the third model contained 11, and the fourth had 14 hidden layers. The PyTorch machine learning programming framework [12] was used to generate all the models. The Neural network consisting of 11 hidden layers showed the best performance. The basic architecture of this network is shown in Fig. 2.

The input layer consists of 14 features. Then the information propagates through 11 hidden layers of varying width. The RELU activation function was used in between

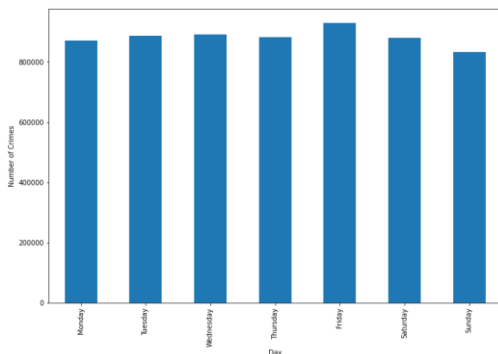


Fig. 3 Distribution of crimes in different days of the week

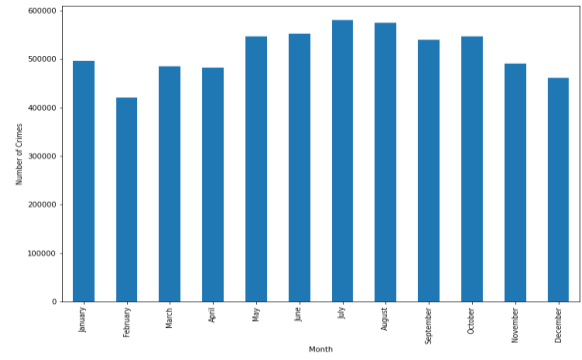


Fig. 4 Distribution of crimes in different month of the year

each hidden layer. Other activation functions were also tried out, but The RELU showed the best accuracy. The layers became wider while going further down towards the middle and then was shrank down to 14 nodes and was finally converged to 2 neurons as the output was a binary classification. The architecture followed a type of double wedge shape with a wide middle with a narrow head and a tail. A SoftMax layer was used at the end to predict the classification label.

A couple of additional data preprocessing was done to help the training of the model and avoid overfitting. The number of data points after the initial preprocessing contained 6.08 million samples where 72% were negative samples and the rest 28% were positive. So, the data was moderately skewed and confused the network, and made it harder to learn. So, a random sampling was done to reduce the data to 2.90 million, where a fraction of the negative samples was discarded. This made the data more uniform and easier to learn with 52% of the new dataset being negative and the rest 48% being positive. Finally, a normalization was applied on all the different features of the dataset by subtracting each value with the mean and dividing it with the standard deviation.

The properties and hyperparameters of the network are shown in table 4. A wide range of different hyperparameter values was tried out with a grid search to pick the most optimum hyperparameters that produced the best results. The Cross-Entropy loss function was used along with an Adam optimizer. 20% of the data was set aside for the testing and the remaining 80% was used for training the model. A batch size of 100 was used while training at a learning rate of 0.01 and the model was trained for 100 epochs. The training accuracy achieved after 100 epochs was about 92% with a 90.77% test accuracy so the model was not overfitted.

IV. DATA VISUALIZATION

When analyzing the dataset, the heatmap shown in Fig. 1 was first used to determine whether crimes were decreasing or increasing. It was found to be gradually going down over the years. Next, the crimes were analyzed per month to observe which months have the highest crime rates, which is shown in Fig. 3. The month of July was determined to be where crimes peak the highest, while February showed the lowest crime rates. Crimes per week were further analyzed, and it was

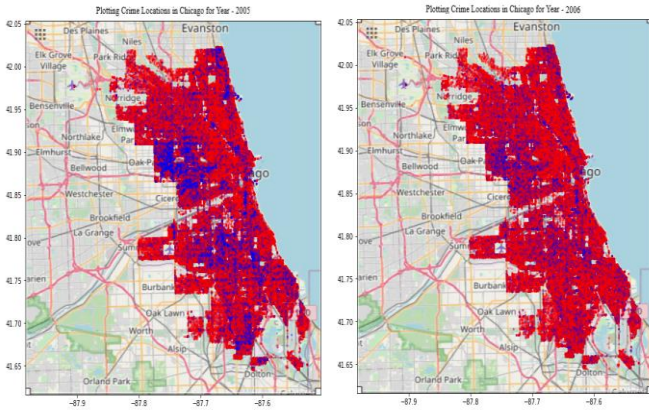


Fig. 5 Comparing 2005 and 2006 locational data based on arrests. Red dots represent crime reports and blue dot represents an arrest was made based on the report.

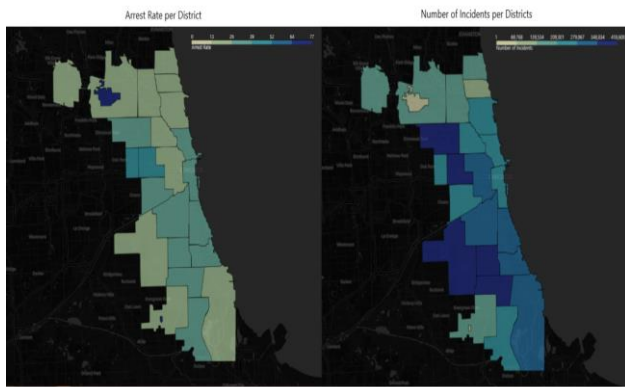


Fig. 6 Comparing the total of arrest rates per district to the total number of incidents per district. Note that the two places with the lowest number of incidents also have the highest arrest rate.

determined that Friday has the highest number of crimes, while Sunday has the lowest number of crimes.

Crimes per location were also analyzed after performing temporal analysis on the dataset. Viewing crimes based on location revealed that the highest number of crimes were committed on the street, followed by crimes committed at residencies, crimes committed at apartments, and crimes committed at sidewalks. Crimes per type were also analyzed, where theft had the highest crime count, followed by battery,

criminal damage, and narcotics.

After data preprocessing and investigation, we noticed a trend with the data based on year. We started plotting out the crime locations, and color coded them based on whether there was an arrest or not for each year. We noticed an increase in arrests in 2004-2005 and then a decrease in arrests made in 2006 shown in Fig. 4. Please note in Fig. 4, we are comparing 2005 and 2006 as this shows the largest change in crime rates in two consecutive years. In Fig. 5, we visualized the Arrest rates per district and the number of incidences per district. This would help point out the districts that need more workforce as there is an obvious lack of resources in these districts needed to keep up with the crime. Please note in Fig. 5, we are comparing the entire data on a district level. We compare the arrest rates with the number of incidents to see if there is a correlation. What we discovered was that some districts with a high arrest rate were due to the low number of incidents, while the districts with the higher number of incidents have a lower arrest rate. Note that there were two districts with a high number of incidents which also have a decent arrest rate. The data visualized in Fig. 5 will help focus down the areas that need to be patrolled more frequently and need more assistance or funding. All visualizations in this section were made using the Folium API.

V. RESULTS

The arrest status was predicted using the preprocessed dataset generated in Section II with the machine learning models described in Section III. Initially, the default parameter values of the Scikit-learn library were used to generate the accuracy parameters. Later, extensive model parameter tuning described in Section III was implemented to increase the models' performance by significant amounts. The corresponding values for the performance metrics of the Random Forest, k-nearest-neighbors, and AdaBoost machine learning algorithms before and after parameter tuning are shown in Table 5. The comparison between the pretuned and tuned accuracies and F1 scores is given in Fig. 6, while that between precision and recall is given in Fig. 7.

The results showed an increase in all three models' overall accuracy after tuning compared to their pre-tuned counterparts with the most significant increase for the Random forest classifier. However, the accuracy of k-NN did not improve much during the tuning process. A large increase in precision

TABLE V
PERFORMANCE OF MODELS WITH DEFAULT AND TUNED PARAMETERS

	Models	Average Accuracy	Average Precision	Average Recall	Average F1 Score
Pre-tuned models with default parameters	Random Forest	84.878%	80.276%	63.826%	70.446%
	k-NN (k = 5)	84.071%	78.756%	59.974%	68.094%
	AdaBoost	85.859%	91.105%	55.791%	68.824%
Model with tuned parameters	Random Forest	88.861%	92.156%	66.283%	77.107%
	k-NN (k = 5)	85.008%	82.825%	59.482%	69.239%
	AdaBoost	87.914%	90.281%	64.344%	75.137%

TABLE VI
PERFORMANCE OF DIFFERENT NEURAL NETWORK MODELS

Number of Layers	Test Accuracy (%)	Precision	Recall	F-1 Score
4	89.97	0.955	0.828	0.887
6	90.43	0.959	0.835	0.893
11	90.77	0.962	0.837	0.895
14	90.31	0.958	0.833	0.891

TABLE VII
PERFORMANCE OF NN MODELS AFTER PARAMETER TUNING

Performance Metric	Results obtained before hyperparameter tuning	Results obtained after hyperparameter tuning
Accuracy	84.971%	90.771%
Precision	89.416%	96.198%
Recall	72.729%	83.706%
F1 Score	80.214%	89.518%

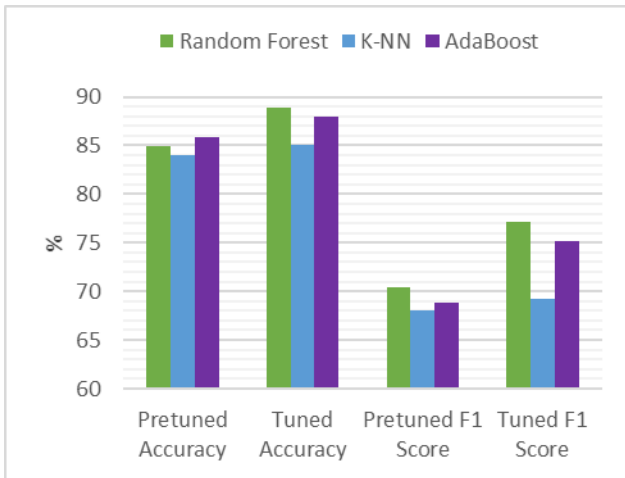


Fig. 7 Comparison between pretuned and tuned accuracies and F1 Scores between the different conventional ML models.

was seen for the Random Forest classifier after tuning, with a significant increase for the k-NN algorithm as well. However, the precision value of the AdaBoost algorithm had in fact decreased after the tuning process. On the other hand, AdaBoost had a considerable enhancement in the recall value, while that of Random Forest and k-NN did not change much due to the parameter tuning. Finally, a significant increase in the F1 score was seen for the Random forest and the AdaBoost algorithm with very little change in that of k-NN after the tuning process. Overall, the Random forest and the AdaBoost showed better performance in terms of the Average accuracy and the F1 score among the three conventional machine learning models.

Finally, four different Neural Network (NN) models with varying architecture were trained on the dataset to predict the police report's arrest status. The hyperparameters of all four neural network architectures trained on the dataset were extensively tuned, and their accuracy parameters were compared to figure out the best model. The results obtained for the different neural network models are given in Table 6. The test accuracy and other performance metrics of the neural

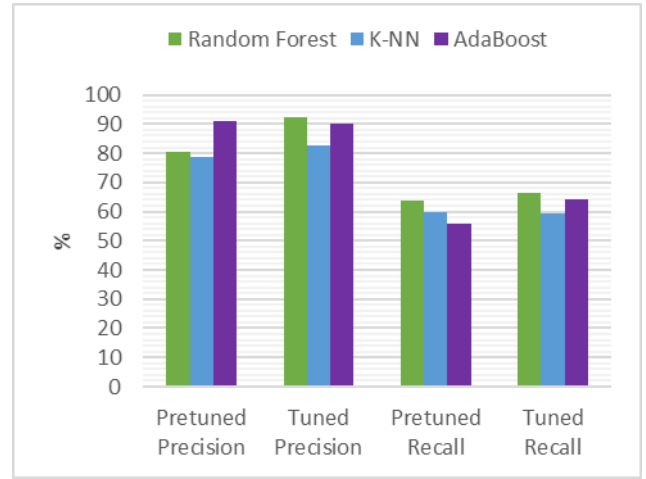


Fig. 8 Comparison between pretuned and tuned precision and recall between the different conventional ML models.

network model before and after hyperparameter tuning are shown in Table 7. A Significant boost in the performance of the model was observed after tuning the hyperparameters. The final test accuracy came up to be 90.77%, which is greater than all the other models that we trained. The Precision, Recall, and F1 score of the neural network is also more impressive compared to the other models.

The comparison between the tuned and final versions of all four machine learning models, namely the Random forest, k-NN, AdaBoost, and neural network are given in Fig. 8. It shows the Neural Network has better performance than the other models in all the different performance metrics. This is especially prominent in the Precision, Recall and F1 scores of the different models.

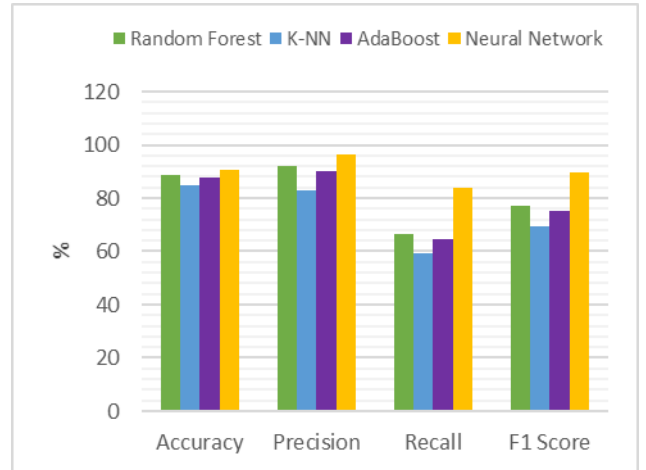


Fig. 9 Comparison between the performance metrics of all four ML Models

VI. CONCLUSION

In this work, we applied K-Nearest Neighbors, AdaBoost, Random Forest, and Neural Network algorithms to form models to forecast future crimes and locations of crimes. Also, we visualized the data using the Folium API for our data breakdown. The neural network based machine learning

model performed the best among the different algorithms achieving an impressive accuracy of 90.77%. It also showed better performance in terms of precision, recall and F-1 scores compared to the other machine learning models. We also implemented several different neural network models with varying architectures and number of layers to determine the best structure for this kind of dataset. This work would help the police department of large metropolitan cities determine which police calls need imminent attention and manage their resources accordingly. It would also help them plan which areas of the city would require more police involvement in the future.

REFERENCES

- [1] S. Kim, P. Joshi, P. S. Kalsi, and P. Taheri, "Crime Analysis Through Machine Learning," in *2018 IEEE 9th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, 1-3 Nov. 2018 2018, pp. 415-420, doi: 10.1109/IEMCON.2018.8614828.
- [2] Y. Lin, T. Chen, and L. Yu, "Using Machine Learning to Assist Crime Prevention," in *2017 6th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI)*, 9-13 July 2017 2017, pp. 1029-1030, doi: 10.1109/IIAI-AAI.2017.46.
- [3] Pratibha, A. Gahalot, Uprant, S. Dhiman, and L. Chouhan, "Crime Prediction and Analysis," in *2nd International Conference on Data, Engineering and Applications (IDEA)*, 28-29 Feb. 2020 2020, pp. 1-6, doi: 10.1109/IDEA49133.2020.9170731.
- [4] A. Jawla, M. Singh, and N. Hooda, "Crime Forecasting using Folium," *Test Engineering and Management*, vol. 82, pp. 16235-16240, 05/31 2020.
- [5] W. Gorr and R. Harries, "Introduction to crime forecasting," *International Journal of Forecasting*, vol. 19, no. 4, pp. 551-555, 2003/10/01/ 2003, doi: [https://doi.org/10.1016/S0169-2070\(03\)00089-X](https://doi.org/10.1016/S0169-2070(03)00089-X).
- [6] M. Granroth-Wilding and S. Clark, "What Happens Next? Event Prediction Using a Compositional Neural Network Model," in *AAAI*, 2016.
- [7] E. Ahishakiye, E. Opiyo, and I. Niyonzima, "Crime Prediction Using Decision Tree (J48) Classification Algorithm," *International Journal of Computer and Information Technology (ISSN: 2279 – 0764)*, 05/15 2017.
- [8] H. Nguyen, C. Cai, and F. Chen, "Automatic classification of traffic incident's severity using machine learning approaches," *IET Intelligent Transport Systems*, vol. 11, 07/14 2017, doi: 10.1049/iet-its.2017.0051.
- [9] J. Cohen, W. L. Gorr, and A. M. Olligschlaeger, "Leading Indicators and Spatial Interactions: A Crime-Forecasting Model for Proactive Police Deployment," *Geographical Analysis*, <https://doi.org/10.1111/j.1538-4632.2006.00697.x> vol. 39, no. 1, pp. 105-127, 2007/01/01 2007, doi: <https://doi.org/10.1111/j.1538-4632.2006.00697.x>.
- [10] A. P. Wheeler and W. Steenbeek, "Mapping the Risk Terrain for Crime Using Machine Learning," *Journal of Quantitative Criminology*, 2020/04/24 2020, doi: 10.1007/s10940-020-09457-7.
- [11] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *the Journal of machine Learning research*, vol. 12, pp. 2825-2830, 2011.
- [12] A. Paszke et al., "Pytorch: An imperative style, high-performance deep learning library," *arXiv preprint arXiv:1912.01703*, 2019.
- [13] *An extensive dataset of crimes in Chicago (2001-2017)*, by City of Chicago. [Online]. Available: <https://www.kaggle.com/currie32/crimes-in-chicago>
- [14] Y.-L. Lin, M.-F. Yen, and L.-C. Yu, "Grid-Based Crime Prediction Using Geographical Features," *ISPRS International Journal of Geo-Information*, vol. 7, no. 8, p. 298, 2018. [Online]. Available: <https://www.mdpi.com/2220-9964/7/8/298>.
- [15] P. Stalidis, T. Semertzidis, and P. Daras, "Examining deep learning architectures for crime classification and prediction," *arXiv preprint arXiv:1812.00602*, 2018.
- [16] Z. M. Wawrzyniak et al., "Data-driven models in machine learning for crime prediction," in *2018 26th International Conference on Systems Engineering (ICSEng)*, 18-20 Dec. 2018 2018, pp. 1-8, doi: 10.1109/ICSENG.2018.8638230.
- [17] M. S. Islam, M. S. Rahman, and M. A. Amin, "Beat Based Realistic Dance Video Generation using Deep Learning," in *2019 IEEE International Conference on Robotics, Automation, Artificial-intelligence and Internet-of-Things (RAAICON)*, 29 Nov.-1 Dec. 2019 2019, pp. 43-47, doi: 10.1109/RAAICON48939.2019.22.
- [18] S. A. Chun, V. A. Paturu, S. Yuan, R. Pathak, V. Atluri, and N. R. Adam, "Crime Prediction Model using Deep Neural Networks," presented at the Proceedings of the 20th Annual International Conference on Digital Government Research, Dubai, United Arab Emirates, 2019. [Online]. Available: <https://doi.org/10.1145/3325112.3328221>.
- [19] A. M. Shermila, A. B. Bellarmine, and N. Santiago, "Crime Data Analysis and Prediction of Perpetrator Identity Using Machine Learning Approach," in *2018 2nd International Conference on Trends in Electronics and Informatics (ICOEI)*, 11-12 May 2018 2018, pp. 107-114, doi: 10.1109/ICOEI.2018.8553904.
- [20] H. K. R. ToppiReddy, B. Saini, and G. Mahajan, "Crime Prediction & Monitoring Framework Based on Spatial Analysis," *Procedia Computer Science*, vol. 132, pp. 696-705, 2018/01/01/ 2018, doi: <https://doi.org/10.1016/j.procs.2018.05.075>.
- [21] N. H. M. Shamsuddin, N. A. Ali, and R. Alwee, "An overview on crime prediction methods," in *2017 6th ICT International Student Project Conference (ICT-ISPC)*, 23-24 May 2017 2017, pp. 1-5, doi: 10.1109/ICT-ISPC.2017.8075335.